Models Performance Review

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1 Coverage

- **Model outputs reviewed:** `OLS`, `OLS3`, `ENET`, `PLS`, `PCR`, `GBRT`, `RF`
- **Prediction horizon reviewed:** 180 monthly out-of-sample periods from **2010-08-31** to **2025-07-31**
- **Full-sample prediction rows per model:** 34,515

2 Executive Takeaways

- **Best predictor on the headline squared-error metric:** `ENET`
- **Best investable long-short model:** `ENET`
- **Best long-only model:** `OLS`
- **Most stable model across rolling windows:** `ENET`
- **Weakest predictor on full-sample R^2:** `RF`
- **Weakest investable model in this batch:** `OLS3`

2.1 My View

1. `ENET` remains the strongest completed model overall. It has the best full-sample out-of-sample R^2, the strongest long-short net Sharpe after 30 bps cost, the highest benchmark information ratio, and the best cumulative long-short path.
2. `OLS` is not the best predictor on R^2, but it remains the strongest long-only model. Its long-only net Sharpe is still the highest in the completed batch even though its R^2 is negative. This again shows that portfolio payoff and point-forecast fit are not the same thing.
3. `PCR` and `PLS` remain credible middle-ground models. They do not beat `ENET`, but both still deliver usable long-short Sharpe above 1 and materially better results than the weaker models.
4. The tree models, `GBRT` and `RF`, are disappointing in the current setup. Neither improves on the linear models. Both are negative on full-sample R^2, weak on long-only and long-short Sharpe, and negative against the benchmark. Additional feature engineering before fitting these models appears necessary for the next batch.
5. `OLS3` remains a baseline only. It is still too restrictive to compete on either predictive fit or economically useful portfolio construction.

3 Headline Comparison

Table 1: Headline comparison across predictive and investable metrics

Model	R2 OOS Full	R2 OOS Large	R2 OOS Small	Mean Rank IC	Long-only Sharpe	Long-short Sharpe	Benchmark IR	LS Ann Ret
ENET	0.0044	0.0039	-0.0003	0.0785	0.970	1.696	0.247	40.3%
PCR	0.0025	0.0004	-0.0014	0.0545	0.825	1.146	0.098	32.8%
PLS	0.0008	-0.0019	-0.0034	0.0724	0.842	1.139	0.141	26.5%
GBRT	-0.0183	-0.0096	-0.0170	0.0573	0.696	0.451	-0.062	26.8%
RF	-0.0235	-0.0150	-0.0102	0.0365	0.378	0.312	-0.067	7.5%
OLS3	-0.0113	-0.0116	-0.0124	0.0306	0.325	-0.054	-0.244	4.4%
OLS	-0.0143	-0.0588	-0.0326	0.1044	1.265	1.301	0.034	-4.5%

Although short selling is prohibited in Vietnam, long-short results are still included for research comparison.

4 Prediction Quality

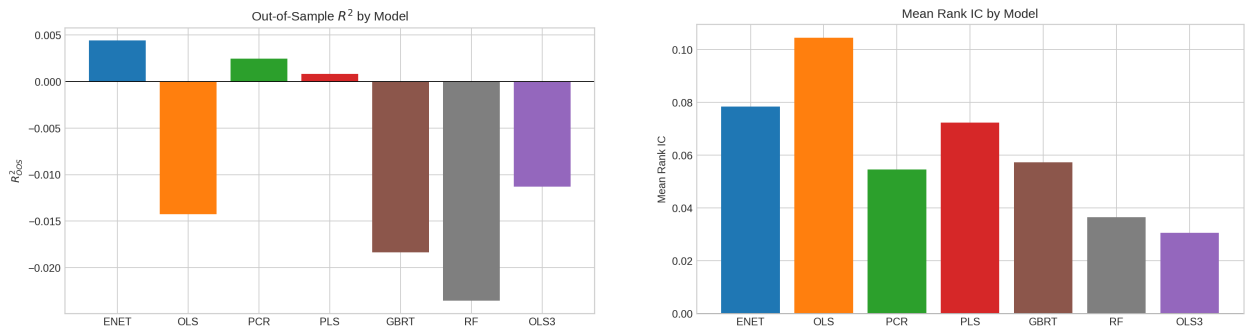

Figure 1: Left: Out-of-sample R^2 by model. Right: Mean rank information coefficient by model.

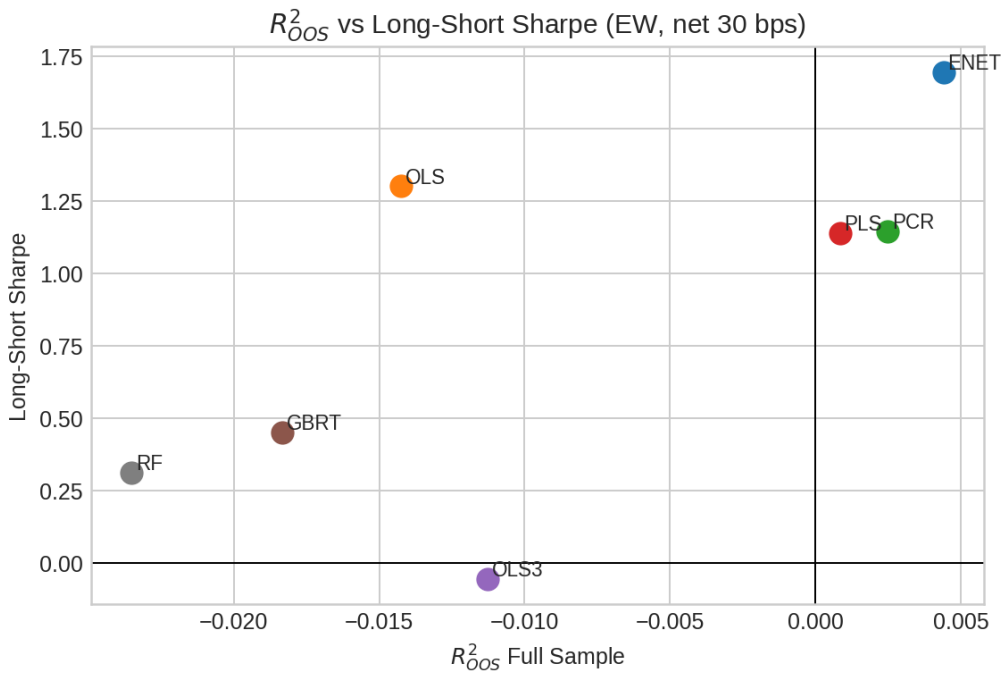

Figure 2: Prediction fit versus long-short Sharpe

4.1 Prediction Metrics Table

Table 2: Prediction-quality metrics

Model	R2 OOS Full	Pearson$(y, \hat{y})$	Spearman$(y, \hat{y})$	Mean Rank IC	Share Positive IC	Sign Accuracy
ENET	0.0044	0.0585	0.0535	0.0785	71.7%	51.5%
OLS	-0.0143	0.0629	0.0850	0.1044	76.7%	53.6%
PCR	0.0025	0.0529	0.0515	0.0545	68.9%	51.1%
PLS	0.0008	0.0551	0.0549	0.0724	70.0%	51.4%
GBRT	-0.0183	-0.0179	-0.0162	0.0573	70.2%	50.8%
RF	-0.0235	-0.0575	-0.0255	0.0365	62.2%	49.6%
OLS3	-0.0113	0.0182	0.0350	0.0306	56.7%	52.7%

Notation definitions.[1]

Interpretation

- `ENET` is the only completed model with clearly positive R^2 in both the full and large-firm samples.
- `PCR` remains positive on the full and large samples, but weaker than `ENET`.

[1]Information coefficient (IC): $IC_t = \text{corr}(\hat{y}_t, y_t)$, where $\hat{y}_t$ are model scores (predicted returns) and y_t are realized returns across the same stocks at date t. Mean IC: Mean IC $= \frac{1}{T}\sum_{t=1}^{T} IC_t$. Information ratio (IR): $AR_t = R_t^{\text{strat}} - R_t^{\text{bench}}$, $IR = \frac{\overline{AR}}{\sigma(AR)}$. Here, R_t^{bench} is the VN30 index proxy (30 largest-cap names in the test period), rebalanced every six months and equal-weighted.

- `PLS` is weakly positive on the full sample and negative on the large and small subsamples.
- `OLS` has the highest mean rank IC and the highest sign accuracy, but negative R^2. This remains the clearest example in the project of why ranking quality and squared-error fit can disagree.
- `GBRT` and `RF` did not add useful predictive power in the current setup.
- `OLS3` remains weak on every predictive dimension reviewed here.

Note: The tree results are negative evidence; more nonlinear structure did not improve performance here.

5 Portfolio Performance

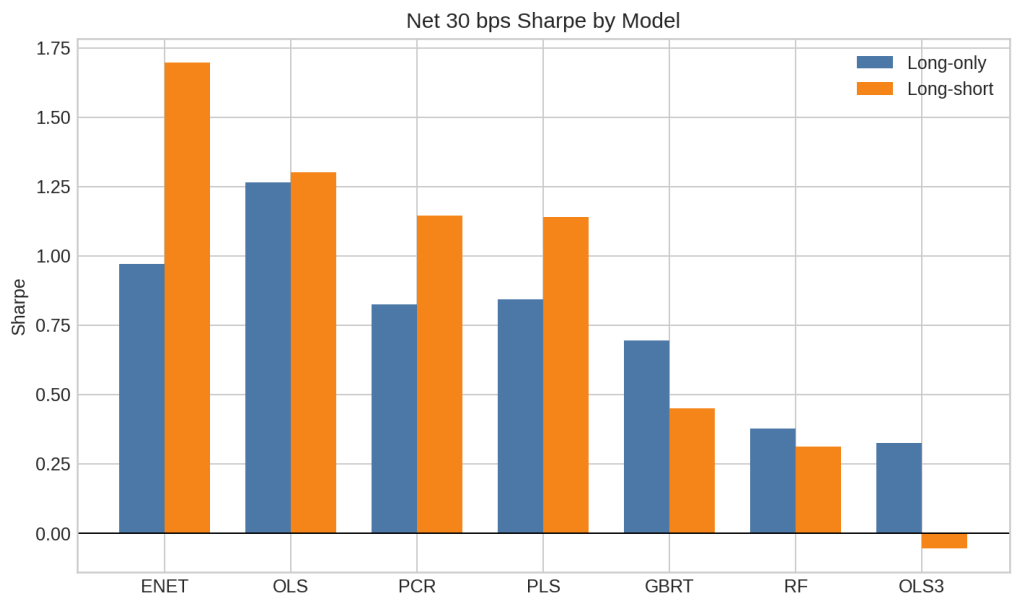

Figure 3: Sharpe ratio comparison by model

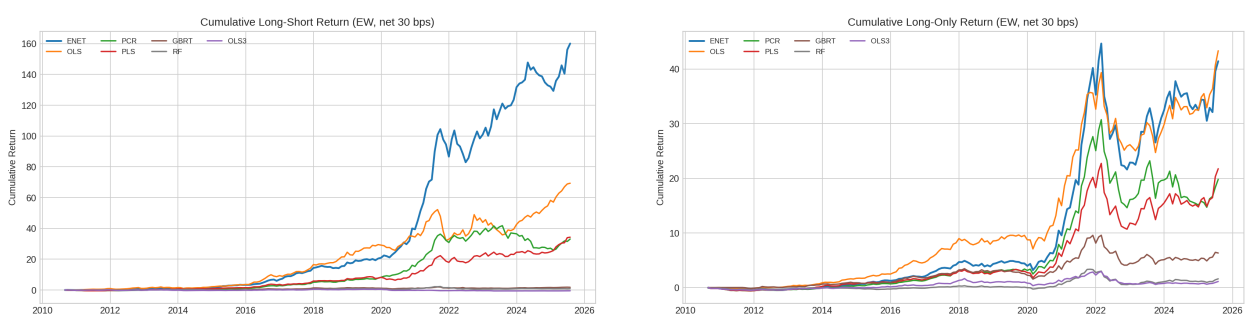

Figure 4: Left: Cumulative long-short return (net 30 bps). Right: Cumulative long-only return (net 30 bps).

5.1 Net 30 bps Portfolio Summary

Table 3: Net 30 bps portfolio performance summary

Model	Long Ann Ret	Long Sharpe	Long Cum Ret (x)	Long Max DD	LS Ann Ret	LS Sharpe	LS Cum Ret (x)	LS Max DD	LS Mean Turnover
ENET	28.4%	0.970	41.434	-51.0%	40.3%	1.696	160.073	-26.2%	1.330
OLS	28.8%	1.265	43.343	-36.3%	32.8%	1.301	69.351	-37.6%	1.639
PCR	22.4%	0.825	19.805	-63.1%	26.5%	1.146	33.100	-50.1%	1.219
PLS	23.2%	0.842	21.752	-57.5%	26.8%	1.139	34.297	-48.1%	1.322
GBRT	15.3%	0.696	6.324	-51.8%	7.5%	0.451	1.767	-40.4%	2.179
RF	6.8%	0.378	1.598	-64.7%	4.4%	0.312	0.859	-53.9%	2.481
OLS3	5.1%	0.325	1.114	-60.2%	-4.5%	-0.054	-0.498	-74.8%	1.055

Interpretation

- `ENET` remains decisively best on the long-short portfolio dimension.
- `OLS` remains the best long-only model in the completed group by net Sharpe and annualized return.
- `PCR` and `PLS` still look investable in research terms, because both produce long-short Sharpe above 1 after cost.
- `GBRT` and `RF` are weak economically; they are not competitive with the stronger linear models.
- `OLS3` remains non-competitive and has the worst long-short drawdown in the completed set.

The chart "R^2 vs portfolio Sharpe" continues to show that the highest-R^2 model and the highest-portfolio model are related but not identical. This is expected in cross-sectional asset pricing:

- R^2 rewards lower squared forecast error,
- the portfolio mainly rewards ranking quality in separating winners from losers.

6 Benchmark Relative Results (Information Ratio)

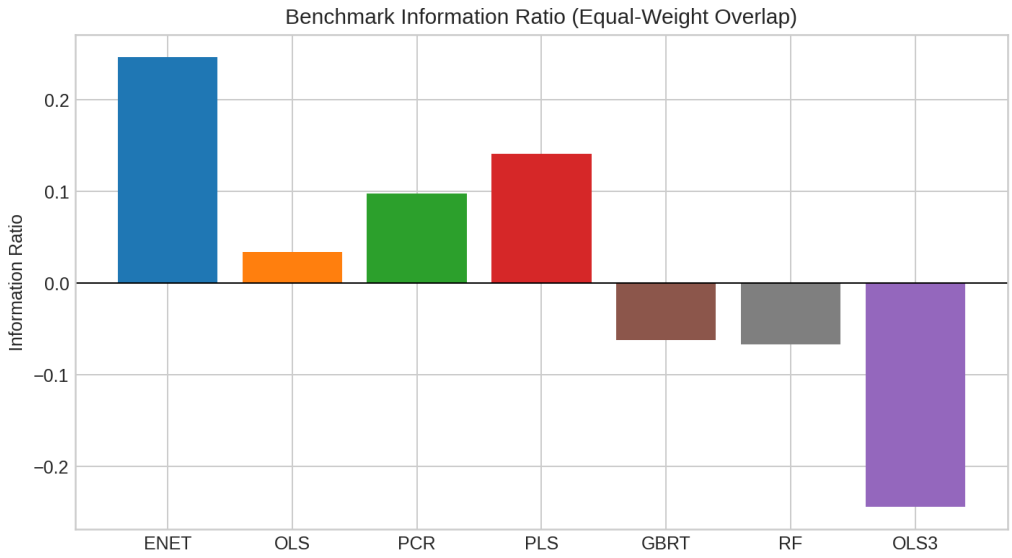

Figure 5: Information ratio relative to equal-weight benchmark

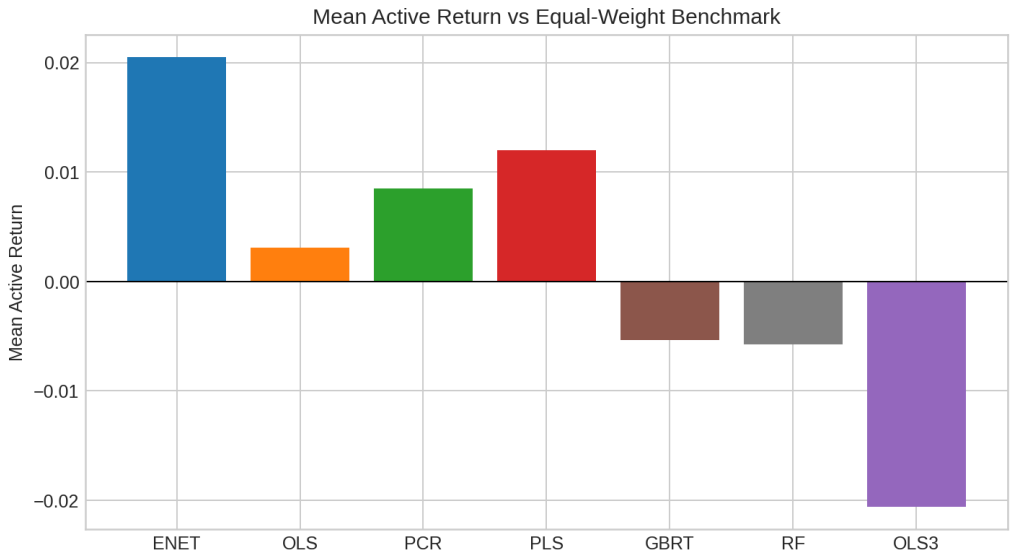

Figure 6: Mean active return relative to equal-weight benchmark

Table 4: Benchmark-relative performance summary

Model	IR vs EW Benchmark	Mean Active Return	Corr vs Benchmark	IR vs VW Benchmark
ENET	0.247	2.05%	0.182	0.329
OLS	0.034	0.31%	-0.043	0.023
PCR	0.098	0.85%	0.090	0.107
PLS	0.141	1.20%	0.098	0.156
GBRT	-0.062	-0.53%	-0.076	-0.032
RF	-0.067	-0.57%	0.046	-0.113
OLS3	-0.244	-2.06%	0.074	-0.223

Interpretation

- `ENET` remains the strongest benchmark-relative strategy by a clear margin.
- `PLS` and `PCR` are second-tier but still positive.
- `OLS` is only slightly positive against the equal-weight benchmark.
- `GBRT` and `RF` are negative against the benchmark, confirming weak economic performance is not only a style effect.
- `OLS3` remains the weakest benchmark-relative model.

7 Rolling Window Stability

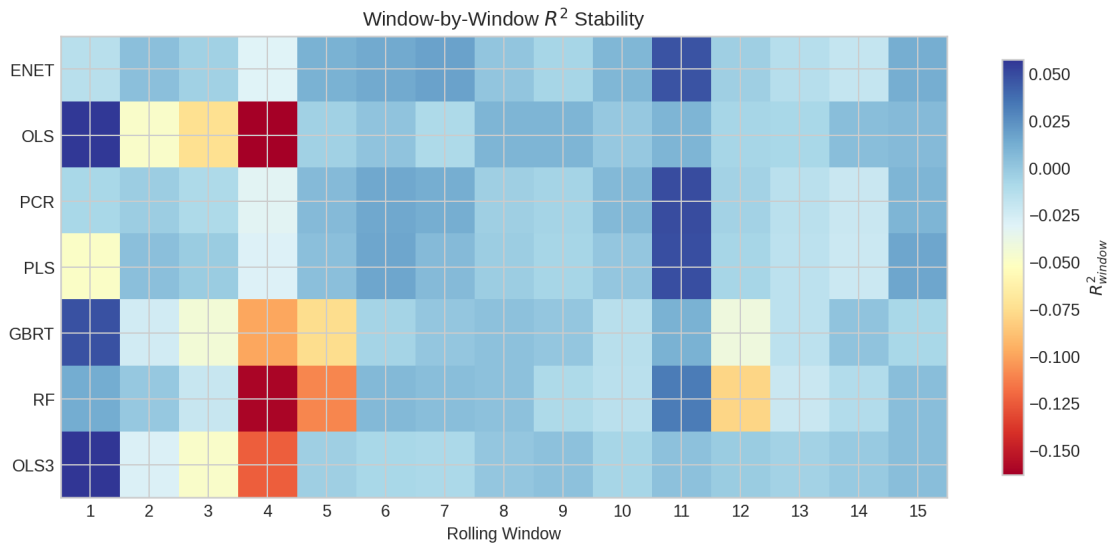

Figure 7: Rolling-window R^2 stability heatmap

Table 5: Rolling-window stability summary

Model	Mean Window R2	Median Window R2	Share Positive Windows	Best Window R2	Worst Window R2
ENET	0.0013	0.0009	53.3%	0.0467	-0.0307
OLS	-0.0149	-0.0011	46.7%	0.0565	-0.1631
PCR	-0.0007	-0.0043	40.0%	0.0492	-0.0326
PLS	-0.0030	-0.0021	40.0%	0.0486	-0.0499
GBRT	-0.0179	-0.0085	33.3%	0.0472	-0.0990
RF	-0.0243	-0.0100	40.0%	0.0322	-0.1597
OLS3	-0.0115	-0.0038	33.3%	0.0574	-0.1242

Interpretation

- `ENET` is the only completed model with positive mean rolling-window R^2 and more than half of windows positive.
- `OLS` has the best single window but materially worse downside window risk than `ENET`.
- `PCR` and `PLS` are middling and slightly negative on average window R^2, but less erratic than `OLS`.
- `GBRT` and `RF` are consistently weak over time, especially `RF` on mean window R^2.
- `OLS3` remains consistently weak, especially on economic metrics.

8 Turnover Cost Sensitivity

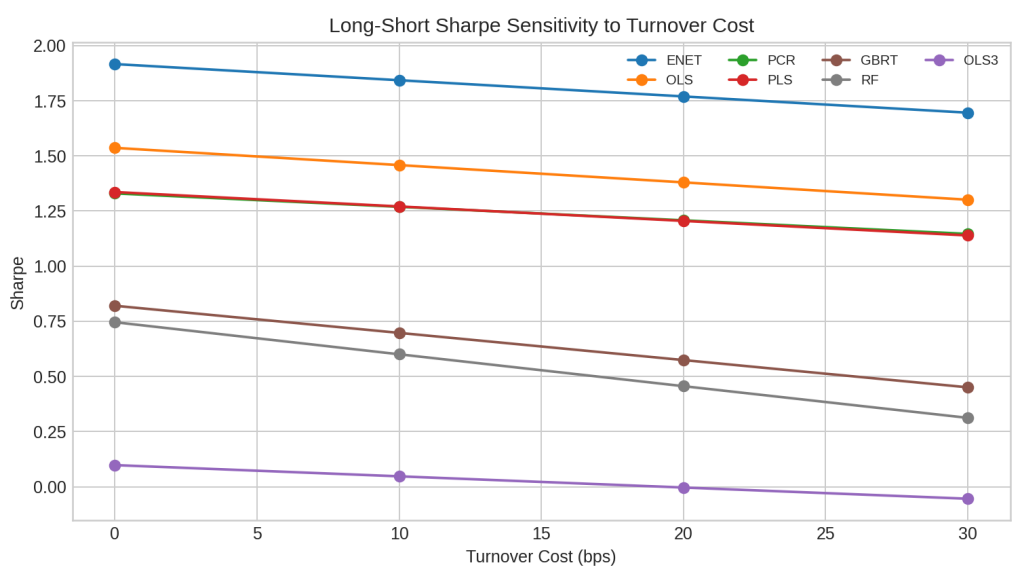

Figure 8: Long-short Sharpe sensitivity to trading-cost assumptions

Table 6: Cost-sensitivity summary

Model	LS Sharpe 0bps	LS Sharpe 10bps	LS Sharpe 20bps	LS Sharpe 30bps	Long Sharpe 30bps
ENET	1.916	1.843	1.770	1.696	0.970
OLS	1.536	1.458	1.380	1.301	1.265
PCR	1.330	1.269	1.208	1.146	0.825
PLS	1.336	1.271	1.205	1.139	0.842
GBRT	0.821	0.698	0.574	0.451	0.696
RF	0.746	0.601	0.456	0.312	0.378
OLS3	0.098	0.047	-0.003	-0.054	0.325

Interpretation

- `ENET` remains the strongest long-short model even after 30 bps turnover cost.
- `OLS` starts from a strong gross result but degrades faster than `ENET` once costs are applied.
- `PLS` and `PCR` are moderately resilient to costs but remain below `ENET`.
- `GBRT` and `RF` are not competitive even before costs; after costs they are clearly second-tier to weak.
- `OLS3` is effectively non-viable once costs are introduced.

9 Feature Importance and Model Shape

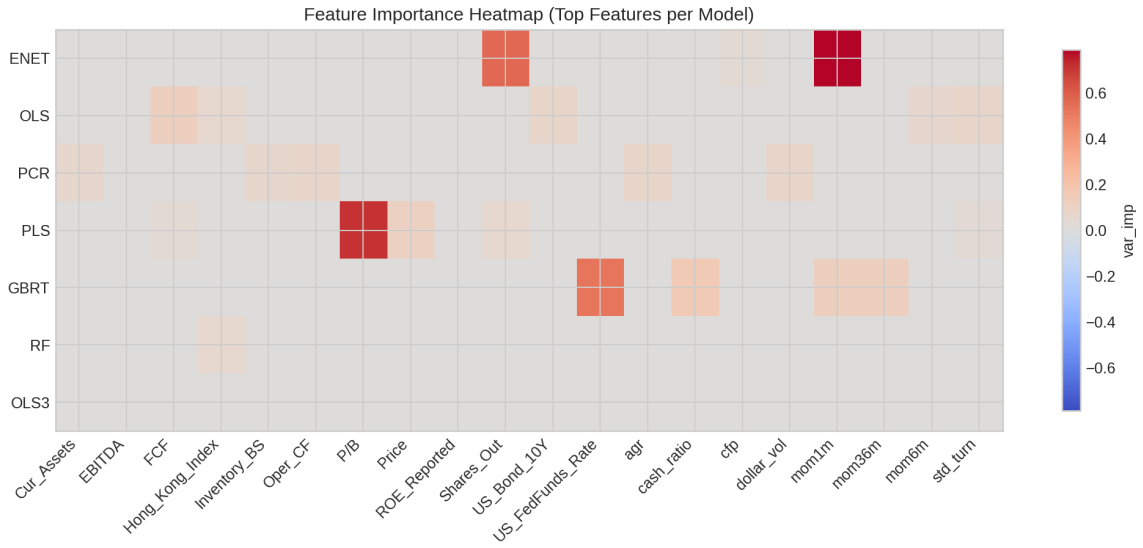

Figure 9: Feature-importance heatmap across models

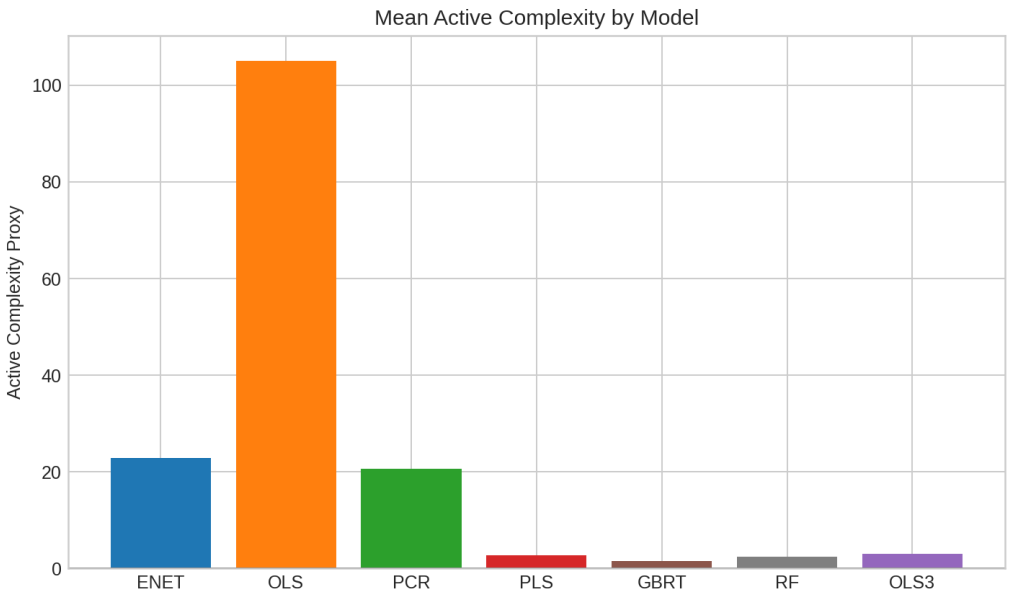

Figure 10: Model complexity comparison

9.1 Top Features by Model

Table 7: Top reported features and normalized importance weights

Model	Top 1	Top 2	Top 3	Top 4	Top 5
ENET	`mom1m` (0.789)	`Shares_Out` (0.566)	`cfp` (0.039)	`ROE_Reported` (0.000)	`EBITDA` (0.000)
OLS	`FCF` (0.120)	`US_Bond_10Y` (0.079)	`std_turn` (0.077)	`mom6m` (0.065)	`Hong_Kong_Index` (0.061)
PCR	`Oper_CF` (0.075)	`agr` (0.075)	`dollar_vol` (0.074)	`Cur_Assets` (0.074)	`Inventory_BS` (0.074)
PLS	`P/B` (0.709)	`Price` (0.107)	`Shares_Out` (0.061)	`FCF` (0.038)	`std_turn` (0.035)
GBRT	`US_FedFunds_Rate` (0.519)	`cash_ratio` (0.159)	`mom36m` (0.122)	`mom1m` (0.119)	`Indonesia_Index` (0.111)
RF	`Indonesia_Index` (0.133)	`Comm_Gold_Spot` (0.073)	`USD_VND_FX` (0.063)	`US_GDP_QoQ` (0.061)	`Hong_Kong_Index` (0.059)
OLS3	`me` (1.215)	`be_me` (-0.004)	`ret_12_1` (-0.212)	–	–

Interpretation: Momentum appears to remain an important factor in predicting the Vietnam market.

9.2 Complexity Summary

Table 8: Complexity diagnostics by model

Model	Mean Validation Score	Mean Active Complexity	Min Active Complexity	Max Active Complexity
ENET	0.141	22.867	5.000	54.000
OLS	0.298	105.000	105.000	105.000
PCR	0.140	20.667	1.000	49.000
PLS	0.141	2.667	1.000	6.000
GBRT	43.208	1.600	1.000	3.000
RF	0.020	2.467	1.000	6.000
OLS3	0.141	3.000	3.000	3.000

Interpretation

- `ENET` remains sparse relative to `OLS` and still concentrates signal on a smaller subset of predictors, led by `mom1m` and `Shares_Out`.
- `OLS` uses the full 105-feature set in every window. This helps explain why it can sort portfolios strongly while remaining unstable on R^2.
- `PLS` remains the most compact latent-factor style model in the completed batch.
- `PCR` is more flexible than `PLS` and can expand to a larger component count in some windows.
- `GBRT` and `RF` both select shallow tree structures on average, consistent with weak out-of-sample results.
- For `GBRT`, some reported feature-importance values are unstable because neutralizing certain features improved fit in the last-window importance routine. These values should be treated as diagnostics rather than strong economic conclusions.
- `OLS3` is simple, but that simplicity does not translate into useful performance in this dataset.

10 Caveats in this version/batch

- Cumulative return figures are large because they are compounded over a long sample and reported from monthly pipeline outputs.
- Absolute economic magnitudes should still be treated with caution. Long-short remains research-only in Vietnam, and current transaction-cost assumptions may still be too light.

A Extra Notes

A.1 How Feature Importance Was Calculated

1. Compute a baseline fit score on `base` using the same `R2OOS` function used elsewhere in the pipeline:

$$R^2_{\text{base}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i y_i^2}$$

2. For each feature f, set that feature to `0.0` in `train`, `validation`, and `base`; refit the model; predict again; and recompute the same `R2OOS`.

3. Measure loss of fit from neutralizing the feature:

$$\Delta R^2_f = R^2_{\text{base}} - R^2_{f=0}$$

4. Normalize these losses into the reported variable-importance share:

$$\text{var_imp}_f = \frac{\Delta R^2_f}{\sum_j \Delta R^2_j}$$

A.2 How Model Complexity Was Calculated

For each rolling window, the model returns:

- `best_params`: hyperparameters chosen on the validation set,
- `best_score`: validation objective value for the selected specification,
- `complexity`: compact model-shape summary.

Complexity by model family

OLS / OLS3

- Implemented with `HuberRegressor` (not classical closed-form OLS).
- `best_score`: validation RMSE.
- `n_nonzero_coef`: number of fitted coefficients with absolute value greater than zero after final `train + validation` refit.

Interpretation: for `OLS`, this is effectively the number of active predictors in the robust linear fit; for `OLS3`, it remains 3 because the model is fixed to `[me, be_me, ret_12_1]`.

ENET

- `best_score`: validation RMSE.
- `best_alpha`: chosen penalty strength.

- `best_l1_ratio`: chosen Elastic Net mixing weight.
- `n_nonzero_coef`: number of coefficients active after shrinkage.

PLS

- `best_score`: validation RMSE.
- `n_components`: chosen latent component count.

PCR

- `best_score`: validation RMSE.
- `n_components`: chosen principal-component count.

GBRT

- `best_score`: validation Huber loss.
- Complexity fields recorded:
  `best_max_depth`, `best_min_samples_split`,
  `best_min_samples_leaf`, `best_learning_rate`.

RF

- `best_score`: validation MSE.
- Complexity fields recorded: `best_max_depth`, `best_max_features`.